As $\|G\| = 3n|D|$, we conclude that

$$|D| = \|G\|/3n \leqslant \delta \cdot 12n = \epsilon n$$

as desired. $\square$

## Exercises

1.$^-$ Show that $K_{1,3}$ is extremal without a $P^3$.

2.$^-$ Given $k > 0$, determine the extremal graphs of chromatic number at most $k$.

3.$^-$ Is there a graph that is edge-maximal without a $K^3$ minor but not extremal?

4. Determine the value of $\mathrm{ex}(n, K_{1,r})$ for all $r, n \in \mathbb{N}$.

5.$^+$ Given $k > 0$, determine the extremal graphs without a matching of size $k$.

   (Hint. Theorem 2.2.3 and Ex. 20, Ch. 2.)

6. Without using Turán's theorem, show that the maximum number of edges in a triangle-free graph of order $n > 1$ is $\lfloor n^2/4 \rfloor$.

7. Show that

$$t_{r-1}(n) \leqslant \tfrac{1}{2} n^2 \, \frac{r-2}{r-1} \,,$$

   with equality whenever $r-1$ divides $n$.

8. Show that $t_{r-1}(n)/\binom{n}{2}$ converges to $(r-2)/(r-1)$ as $n \to \infty$.

   (Hint. $t_{r-1}\big((r-1)\lfloor \frac{n}{r-1} \rfloor\big) \leqslant t_{r-1}(n) \leqslant t_{r-1}\big((r-1)\lceil \frac{n}{r-1} \rceil\big)$.)

9. Does every large enough graph $G$ with at most $c\,|G|$ edges, where $c$ is any constant, contain a set of 100 independent vertices?

10. Show that deleting at most $(m-s)(n-t)/s$ edges from a $K_{m,n}$ will never destroy all its $K_{s,t}$ subgraphs.

11. For $0 < s \leqslant t \leqslant n$ let $z(n,s,t)$ denote the maximum number of edges in a bipartite graph whose partition sets both have size $n$, and which does not contain a $K_{s,t}$. Show that $2\,\mathrm{ex}(n, K_{s,t}) \leqslant z(n,s,t) \leqslant \mathrm{ex}(2n, K_{s,t})$.

12.$^+$ Let $1 \leqslant r \leqslant n$ be integers. Let $G$ be a bipartite graph with bipartition $\{A, B\}$, where $|A| = |B| = n$, and assume that $K_{r,r} \not\subseteq G$. Show that

$$\sum_{x \in A} \binom{d(x)}{r} \leqslant (r-1)\binom{n}{r}.$$

   Using the previous exercise, deduce that $\mathrm{ex}(n, K_{r,r}) \leqslant c n^{2-1/r}$ for some constant $c$ depending only on $r$.

13. The *upper density* of an infinite graph $G$ is the lim sup of the maximum edge densities of its (finite) $n$-vertex subgraphs as $n \to \infty$.

    (i) Show that, for every $r \in \mathbb{N}$, every infinite graph of upper density $> \frac{r-2}{r-1}$ has a $K^r_s$ subgraph for every $s \in \mathbb{N}$.

    (ii) Deduce that the upper density of infinite graphs can only take the countably many values of $0, 1, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots$.

14. Given a tree $T$, find an upper bound for $\mathrm{ex}(n, T)$ that is linear in $n$ and independent of the structure of $T$, i.e. depends only on $|T|$.

15. Show that the Erdős-Sós conjecture is best possible in the sense that, for every $k$ and infinitely many $n$, there is a graph on $n$ vertices and with $\frac{1}{2}(k-1)n$ edges that contains no tree with $k$ edges.

16.$^-$ Prove the Erdős-Sós conjecture for the case when the tree considered is a star.

17. Prove the Erdős-Sós conjecture for the case when the tree considered is a path.

    (Hint. Use Exercise 9 of Chapter 1.)

18. Can large average degree force the chromatic number up if we exclude some tree as an induced subgraph? More precisely: For which trees $T$ is there a function $f\colon \mathbb{N} \to \mathbb{N}$ such that, for every $k \in \mathbb{N}$, every graph of average degree at least $f(k)$ either has chromatic number at least $k$ or contains an induced copy of $T$?

19. Given two numerical graph invariants $i_1$ and $i_2$, write $i_1 \leqslant i_2$ if we can force $i_2$ to be arbitrarily high on some subgraph of $G$ by assuming that $i_1(G)$ is large enough. (Formally: write $i_1 \leqslant i_2$ if there exists a function $f\colon \mathbb{N} \to \mathbb{N}$ such that, given any $k \in \mathbb{N}$, every graph $G$ with $i_1(G) \geqslant f(k)$ has a subgraph $H$ with $i_2(H) \geqslant k$.) If $i_1 \leqslant i_2$ as well as $i_1 \geqslant i_2$, write $i_1 \sim i_2$. Show that this is an equivalence relation for graph invariants, and sort the following invariants into equivalence classes ordered by $<$: minimum degree; average degree; connectivity; arboricity; chromatic number; colouring number; choice number; $\max\{\, r \mid K^r \subseteq G \,\}$; $\max\{\, r \mid TK^r \subseteq G \,\}$; $\max\{\, r \mid K^r \preccurlyeq G \,\}$; $\min \max d^+(v)$, where the maximum is taken over all vertices $v$ of the graph, and the minimum over all its orientations.

20.$^+$ Prove, from first principles and without using average or minimum degree arguments, the existence of a function $f\colon \mathbb{N} \to \mathbb{N}$ such that every graph of chromatic number at least $f(r)$ has a $K^r$ minor.

    (Hint. Use induction on $r$. For the induction step $(r-1) \to r$ try to find a connected set $U$ of vertices whose neighbours induce a subgraph that needs enough colours to contract to $K^{r-1}$. If no such set $U$ exists, show that the given graph can be coloured with fewer colours than assumed.)

21. Given a graph $G$ with $\varepsilon(G) \geqslant k \in \mathbb{N}$, find a minor $H \preccurlyeq G$ such that $\delta(H) \geqslant k \geqslant |H|/2$.

22.$^{+}$ Find a constant $c$ such that every graph with $n$ vertices and at least $n + 2k(\log k + \log\log k + c)$ edges contains $k$ edge-disjoint cycles (for all $k \in \mathbb{N}$). Deduce an edge-analogue of the Erdős-Pósa theorem (2.3.2).

(Hint. Assuming $\delta \geqslant 3$, delete the edges of a short cycle and apply induction. The calculations are similar to the proof of Lemma 2.3.1.)

23. Simplify the proof of Theorem 7.2.3 by using Exercise 36, Ch. 3.

24.$^{+}$ Show that any function $h$ as in Lemma 3.5.1 satisfies the inequality $h(r) > \frac{1}{8}r^2$ for all even $r$, and hence that Theorem 7.2.3 is best possible up to the value of the constant $c$.

25. Characterize the graphs with $n$ vertices and more than $3n - 6$ edges that contain no $TK_{3,3}$. In particular, determine $\mathrm{ex}(n, TK_{3,3})$.

(Hint. You may use the theorem of Wagner that every edge-maximal graph without a $K_{3,3}$ minor can be constructed recursively from maximal planar graphs and copies of $K^5$ by pasting along $K^2$s.)

26.$^{-}$ Derive the four colour theorem from Hadwiger's conjecture for $r = 5$.

27.$^{-}$ Show that Hadwiger's conjecture for $r + 1$ implies the conjecture for $r$.

28.$^{-}$ Deduce the following weakening of Hadwiger's conjecture from known results: given any $\epsilon > 0$, every graph of chromatic number at least $r^{1+\epsilon}$ has a $K^r$ minor, provided that $r$ is large enough.

29.$^{-}$ Show that any graph constructed as in Proposition 7.3.1 is edge-maximal without a $K^4$ minor.

30. Prove the implication $\delta(G) \geqslant 3 \Rightarrow G \supseteq TK^4$.

(Hint. You may use any result from Section 7.3.)

31. A multigraph is called *series-parallel* if it can be constructed recursively from a $K^2$ by the operations of subdividing and of doubling edges. Show that a 2-connected multigraph is series-parallel if and only if it has no (topological) $K^4$ minor.

32. Without using Theorem 7.3.8, prove Hadwiger's conjecture for all graphs of girth at least 11 and $r$ large enough. Without using Corollary 7.3.9, show that there is a constant $g \in \mathbb{N}$ such that all graphs of girth at least $g$ satisfy Hadwiger's conjecture, irrespective of $r$.

33.$^{+}$ Prove Hadwiger's conjecture for $r = 4$ from first principles.

34.$^{+}$ Prove Hadwiger's conjecture for line graphs.

35. Prove Corollary 7.3.5.

36.$^{-}$ In the definition of an $\epsilon$-regular pair, what is the purpose of the requirement that $|X| \geqslant \epsilon\,|A|$ and $|Y| \geqslant \epsilon\,|B|$?

37.$^{-}$ Show that any $\epsilon$-regular pair in $G$ is also $\epsilon$-regular in $\overline{G}$.

38. Does the regularity lemma for $m = 1$ imply the general version?

39. Consider a partition of a finite set $V$ into $k$ equally sized subsets. Show that the complete graph on $V$ has about $k-1$ as many edges between different partition sets as edges inside partition sets. Explain how this leads to the choice of $m := 1/\gamma$ in the proof of the Erdős-Stone theorem.

40. (i) Deduce the regularity lemma from the assumption that it holds, given $\epsilon > 0$ and $m \geqslant 1$, for all graphs of order at least some $n = n(\epsilon, m)$.

(ii) Prove the regularity lemma for sparse graphs – more precisely, for every sequence $(G_n)_{n \in \mathbb{N}}$ of graphs $G_n$ of order $n$ such that $\|G_n\|/n^2 \to 0$ as $n \to \infty$.

41.$^-$ Consider a pair $(A, B)$ of disjoint vertex sets in a graph, and a pair of subsets $X \subseteq A$ and $Y \subseteq B$ that are small in that $|X||Y| \leqslant \epsilon\,|A||B|$. For which $d$ does $(X, Y)$ satisfy the requirement of $(\epsilon, d)$-uniformity for $(A, B)$?

42. Compare the definitions of $\epsilon$-regularity and $\epsilon$-uniformity for pairs of disjoint vertex sets in a graph:

(i)$^-$ Show that every $\epsilon$-regular pair is also $\epsilon$-uniform.

(ii) Show that every $\epsilon$-uniform pair is $\sqrt[3]{\epsilon}$-regular.

43. Discuss the factor of $\binom{k}{2}$ in Proposition 7.6.1. Is it natural? Is it important? Why is it there?

44. Deduce Proposition 7.6.1 from Lemma 7.6.2.

45. Replace the use of the blow-up lemma with the counting lemma...

(i) ...in the proof of Theorem 7.1.2;

(ii) ...in the proof of Theorem 9.2.2.

46. A consequence of the Erdős-Stone theorem is that $\gamma n^2$ more edges than are needed to force a $K^r$ subgraph will force many such subgraphs, as many as a $K^r_s$ contains. Can you prove this directly using the removal lemma? Can you find even more $K^r$ subgraphs in this way?

47. Let $H$ be any graph, of order $k$ say. Does the removal lemma say, essentially, that we can hit all the copies of $H$ in arbitrary graphs $G$ by $o(n^2)$ edges, where $n := |G|$, if $H$ is such that no more than $o(n^k)$ of the $n^k$ maps $V(H) \to V(G)$ induce homomorphisms $H \to G$? If not, what is missing?

48.$^-$ Find a class of 3-connected planar graphs over which triangles are spread thinly.

49. (i) In the definition of 'thinly spread', and again in Corollary 7.6.5, there is a condition requiring that every edge of $G$ lie in at least one copy of $H$. If we drop this assumption, will Theorem 7.6.4 and Corollary 7.6.5 fail?

(ii) In our earlier informal discussion of a typical scenario for the removal lemma, which begins with the inequalities $(*)$ and concludes that many copies of $H$ are lumped together at just a few edges of $G$, there is no such condition as in (i). Why was it not needed there?

50. In the definition of 'thinly spread', weaken the condition that every edge of $G$ lies in at least one copy of $H$ to the condition that the edges of $G$ lie in at least $\beta$ copies of $H$ on average, for some $\beta > 0$ that is the same for all $G \in \mathcal{G}$. Does Theorem 7.6.4 still hold?

51. We saw that Theorem 7.7.2 implies Theorem 7.7.1. Does the converse implication hold, too?

# Notes

The standard reference work for results and open problems in extremal graph theory (in a very broad sense) is still B. Bollobás, *Extremal Graph Theory*, Academic Press 1978. A kind of update on the book is given by its author in his chapter of the *Handbook of Combinatorics* (R.L. Graham, M. Grötschel & L. Lovász, eds.), North-Holland 1995. An instructive survey of extremal graph theory in the narrower sense of Section 7.1 is given by M. Simonovits in (L.W. Beineke & R.J. Wilson, eds.) *Selected Topics in Graph Theory 2*, Academic Press 1983. This paper focuses among other things on the particular role played by the Turán graphs. A more recent survey by the same author can be found in (R.L. Graham & J. Nešetřil, eds.) *The Mathematics of Paul Erdős*, Vol. 2, Springer 1996.

Turán's theorem is not merely one extremal result among others: it is the result that sparked off the entire line of research. Our first proof of Turán's theorem is essentially the original one; the second is a version of a proof of Zykov due to Brandt.

Turán's theorem has been generalized as follows. Suppose that, for some fixed $r \geqslant 3$, we wish to construct a graph on $n$ vertices with at least $\gamma n^2$ edges, where now $\frac{1}{2}\frac{r-2}{r-1} < \gamma < \frac{1}{2}$, in such a way as to create as few $K^r$ subgraphs as possible. The *clique density theorem* says that, for fixed $\gamma$, the asymptotically best way to do this is to form a complete multipartite graph in which all classes have the same size except for one, which may be smaller. How many such classes there are depends on $\gamma$, but not on $n$: as in Turán's theorem, $s$ classes will always give about $\gamma n^2$ edges for $\gamma = \frac{1}{2}\frac{s-1}{s}$. The clique density theorem had been conjectured by Lovász and Simonovits in 1983, and was finally proved for all $r$ by C. Reiher, The clique density theorem, *Ann. Math.* **184** (2016), 683–707, arXiv:1212.2454.

Our version of the Erdős-Stone theorem is a slight simplification of the original. A direct proof, not using the regularity lemma, is given in L. Lovász, *Combinatorial Problems and Exercises* (2nd edn.), North-Holland 1993. Its most fundamental application, Corollary 7.1.3, was only found 20 years after the theorem, by Erdős and Simonovits (1966).

Of our two bounds on $\mathrm{ex}(n, K_{r,r})$ the upper one is thought to give the correct order of magnitude. For vastly off-diagonal complete bipartite graphs this was verified by J. Kollár, L. Rónyai & T. Szabó, Norm-graphs and bipartite Turán numbers, *Combinatorica* **16** (1996), 399–406, who proved that $\mathrm{ex}(n, K_{r,s}) \geqslant c_r n^{2-\frac{1}{r}}$ when $s > r!$.

Details about the Erdős-Sós conjecture, including an approximate solution for large $k$, can be found in the survey by Komlós and Simonovits cited below. The case where the tree $T$ is a path (Exercise 17) was proved by